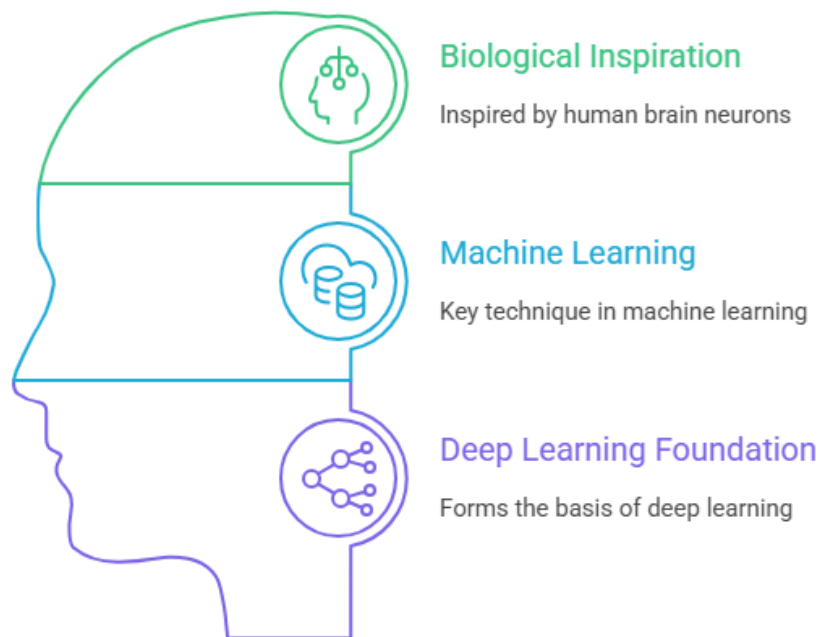


# Understanding Neural Networks: Concepts & Fundamentals

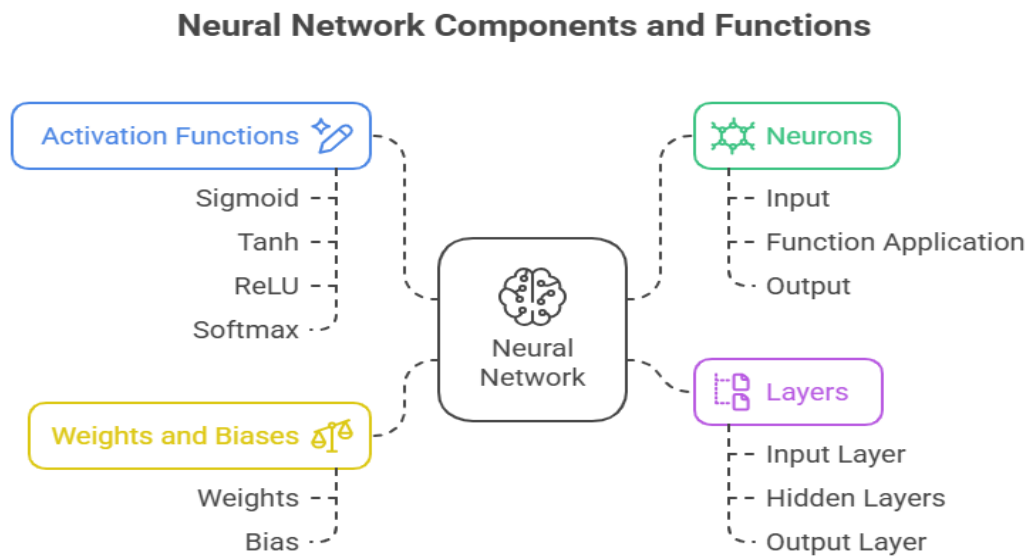
## Introduction

- A Neural Network is a computational model inspired by the way biological neurons in the human brain process information.
- It is a key technique in Machine Learning and forms the foundation of Deep Learning.

## Understanding Neural Networks



## Key Components of a Neural Network



### 1. Neurons (Nodes)

- Basic processing units that take input, apply a function, and pass output forward.

### 2. Layers

- Input Layer: Accepts raw data.
- Hidden Layers: Perform computations and feature extraction.
- Output Layer: Produces the final prediction or classification.

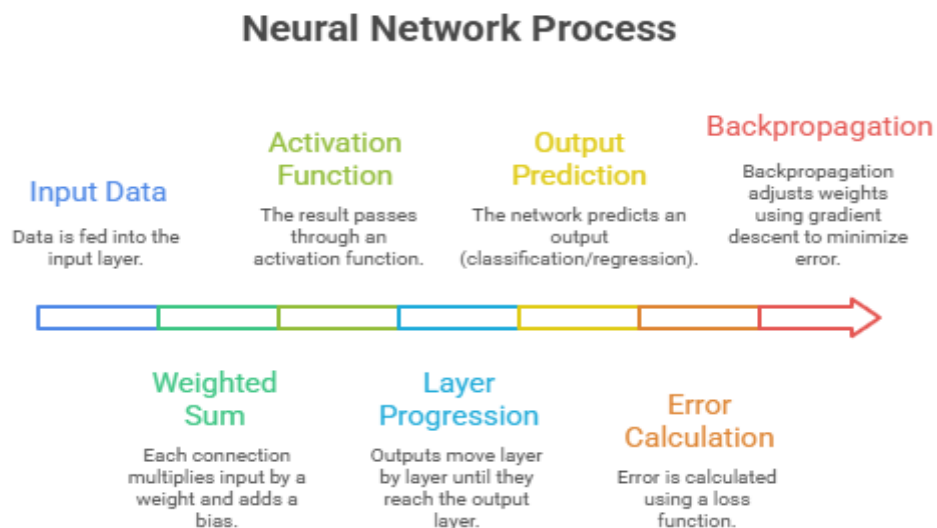
### 3. Weights and Biases

- Weights: Parameters that define the strength of connections between neurons.
- Bias: Helps shift the activation function for better learning.

#### 4. Activation Functions

- Decide whether a neuron should be activated or not.
- Examples: Sigmoid, Tanh, ReLU, Softmax.

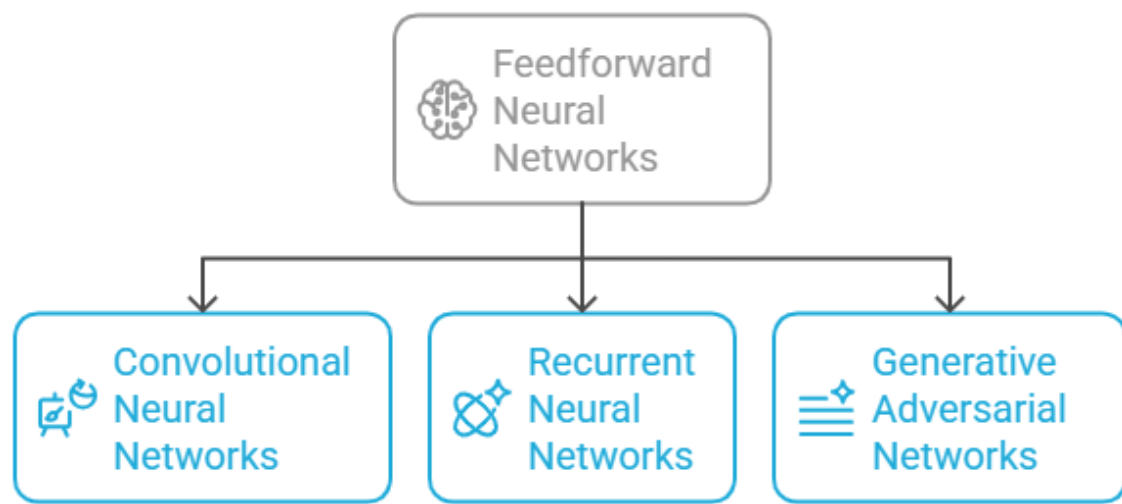
#### How Neural Networks Work (Step-by-Step)



1. Input data is fed into the input layer.
2. Each connection multiplies the input by a weight and adds a bias.
3. The result passes through an activation function.
4. Outputs move layer by layer until they reach the output layer.
5. The network predicts an output (classification/regression).

6. Error is calculated using a loss function.
7. Backpropagation adjusts weights using gradient descent to minimize error.

## Types of Neural Networks



1. Feedforward Neural Networks (FNN) – Basic architecture, data flows in one direction.
2. Convolutional Neural Networks (CNN) – Best for image recognition.
3. Recurrent Neural Networks (RNN) – Best for sequential data (text, speech).
4. Generative Adversarial Networks (GANs) – Used for generating new data (images, audio).

## Advantages

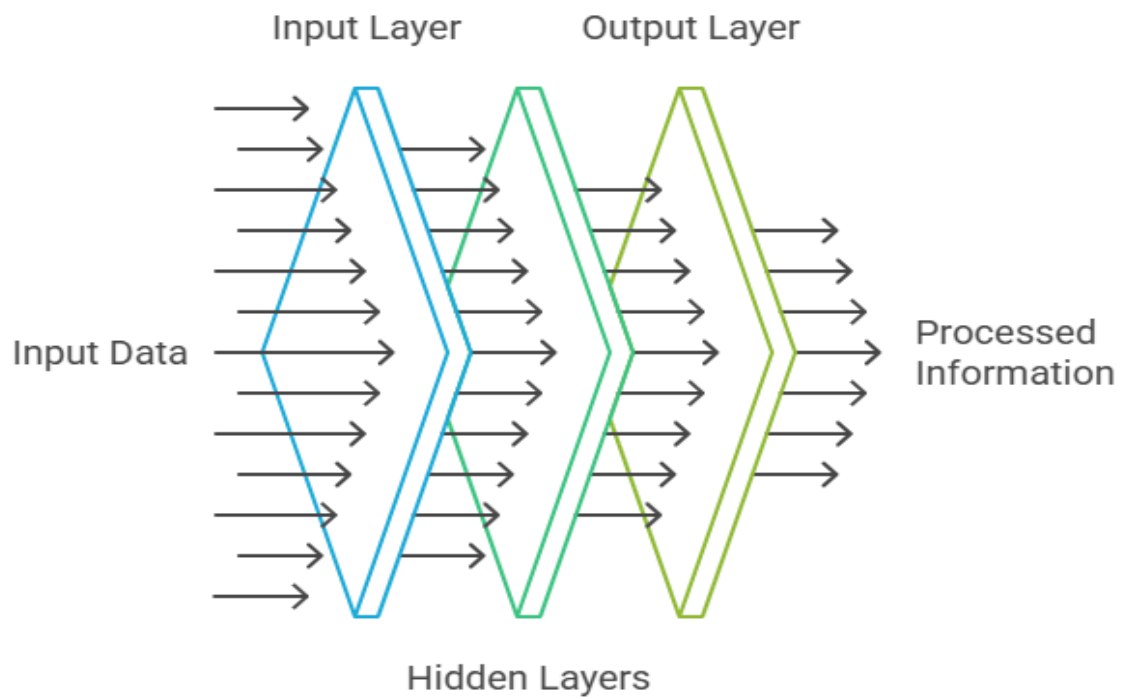
- Can model complex relationships.
- Automatically extract features (Deep Learning).
- Powerful for images, speech, and NLP tasks.

## Challenges

- Requires large datasets.
- Computationally expensive.
- Risk of overfitting.
- Less interpretable ("black box").



## Visual Representation



- Diagram showing Input Layer → Hidden Layers → Output Layer
- Highlight flow of data & activation

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